Program: **Diploma in Food Processing Technology**	
Course Code: **4423**	Course Title: **Food Preservation**
Semester: **4**	Credits: **4**
Course Category: **Program Core**	
Periods per week: **4 (L:3 T:1 P:0)**	Periods per semester: **60**

Course Objectives:

1. To understand the basic concepts and principles of food preservation.
2. To study high and low temperature preservation techniques
3. To impart knowledge about genetically modified foods, its health concerns and implications

Course Prerequisites:

Topic	Program/Course Name
Basic knowledge of food processing technology	Food processing technology

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the basic concepts and principles of food preservation.	14	Understanding
CO2	Identify various high and low temperature preservation methods	16	Understanding
CO3	Explain the mechanism involved in processing and preservation of different food groups	14	Understanding
CO4	Review the concept and safety concerns of Genetically modified foods.	16	Understanding

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Illustrate the basic concepts and principles of food preservation**.		
M1.01	Discuss the basic principles of food preservation	5	Understanding
M1.02	Summarize Scope of food processing;	5	Understanding
M1.03	Describe Food additives of food products	4	Understanding
Contents: Basic principles of food preservation. Scope of food processing; Food additives: definition, types and functions, permissible limits and safety aspects.			
CO2	**Identify various high and low temperature preservation methods**		
M2.01	Recognize Processing and preservation by heat – blanching, pasteurization, sterilization	4	Understanding
M2.02	Recognize Processing and preservation by low-temperature-refrigeration, freezing, and dehydro-freezing	4	Understanding
M2.03	Discuss about canning	4	Understanding
M2.04	Understand microwave heating, baking, roasting and frying.	3	Understanding
	Series Test –I	1	

Contents: Processing and preservation by heat – blanching, pasteurization, sterilization, Processing and preservation by low-temperature- refrigeration, freezing, and dehydro-freezing, canning, microwave heating, baking, roasting and frying			
CO3	**Explain the mechanism involved in processing and preservation of different food groups**		
M3.01	Discuss Processing and preservation by drying, concentration and evaporation	5	Understanding
M3.02	Summarize the types of dryers	5	Understanding
M3.03	Outline Non-thermal methods - irradiation, high pressure, hurdle technology	4	Understanding
Contents: Processing and preservation by drying, concentration and evaporation-types of dryers, non-thermal methods, irradiation, high pressure, hurdle technology.			
CO4	**Review the concept and safety concerns of Genetically modified foods.**		
M4.01	Review Genetically Modified Foods (GMF) and their health implications	5	Understanding
M4.02	Understand safety concerns of genetically modified foods	5	Understanding
M4.03	Explain Nutrigenomics	5	Understanding
	Series Test–II	1	
contents: Genetically Modified Foods (GMF) and their health implications, safety concerns of genetically modified foods; Nutrigenomics - interaction between gene-diet-disease, benefits and risk;			

Text/Reference

T/R	Book Title/Author
R1	Potter NN & Hotchkiss 1997. Food Science. 5th Ed. CBS.
R2	Fellows PJ. 2005. Food Processing Technology: Principle and Practice. 2nd Ed. CRC
R3	Ramaswamy H & Marcotte M. 2006. Food Processing: Principles and Applications. Taylor & Francis.

R4	Food Processing: Principles and Applications, J. Scott Smith, Y.H. Hui, Wiley India Pvt Ltd (28 October 2013).
R5	Manay, N. S. and Shadaksharaswamy, 1997, Foods Facts and Principles, New Age international (P) Ltd., Publishers New Delhi

Online Resources

Sl.No	Website Link
1	https://www.hiperbaric.com/en/hpp-technology/what-is-hpp/
2	https://www.webmd.com/diet/genetically-modified-foods-overview
3	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2137135/#:~:text=Nutrigenomics%20therefore%20initially%20referred%20to,protect%20the%20genome%20from%20damage.
4	https://www.foodnetwork.com/how-to/packages/food-network-essentials/what-is-blanching
5	https://www.britannica.com/topic/canning-food-processing